Paleomagnetic Properties of Some of the Precambrian Rocks in Missouri

I-CHI HSU, R. E. ANDERSON,[1] AND LEROY SCHARON

Department of Earth Sciences
Washington University, St. Louis, Missouri

Measurements of remanent magnetization were made on cores prepared from 54 samples collected from a 50 mi² area of the St. Francois Mountains, Missouri. The rocks studied are devitrified Precambrian volcanics. In some of these rocks, the stable magnetization has high coercivity and appears to reside in fine-grained magnetite that was formed by devitrification and vapor-phase crystallization. This magnetization is inferred to be chemical remanent magnetization. The remanent magnetization directions from all the samples are closely grouped after demagnetization in ac fields of 500 to 800 oersteds. The calculated mean direction is coincident with the mean directions of other Precambrian igneous rocks of the St. Francois Mountains. The calculated pole position is 141.0°W, 0.9°S.

INTRODUCTION

The paleomagnetism of igneous rocks in the St. Francois Mountains in Missouri was first investigated in 1959 by the Department of Earth Sciences of Washington University. Some of the early results were used by *Allingham* [1960] as an aid in the interpretation of aeromagnetic data associated with these rocks. Paleomagnetic results on most of the Precambrian rocks were made available by *Hays* [1961] and by *Hays and Scharon* [1966]. *Anderson* [1962], as part of this general paleomagnetic program, mapped a thick layered sequence of pyroclastic flows and related volcanic rocks southwest of Ironton, Missouri. This paper presents an investigation of the paleomagnetic properties of this volcanic group.

GEOLOGY

In general, the Precambrian rocks within the sampled area comprise a sequence of rhyolitic and andesitic volcanics that are intruded by gabbro and granite porphyry (Figure 1). Field and petrographic studies indicate that most of the volcanic rocks formed by explosive eruptions of ash and pumice are ash-flow tuff sheets (terminology used here is after *Smith* [1960]). Minor amounts of bedded tuff and lava are also present.

Most of the rocks contain planar flow or compaction structures. Throughout most of the area the planar structures as well as the unit contacts have gentle dips. Chaotic flow folding is common in the lava flows and in some of the ash-flow tuffs, especially in the upper part of the stratigraphic section. In no part of the area, not even at the margins of intrusive bodies, were high dip values recorded which could possibly be related to post-volcanic tectonic activity. Essentially all of the rock structures and textures are primary, and the principal configuration of the regional structures is that which developed during the volcanic epoch.

Most of the extrusive rocks contain iron oxide phenocrysts. Studies with the microscope revealed complex exsolution features in these grains. In general, many of the euhedral magnetite grains contain lamellas and blebs of exsolved ilmenite, and the iron fraction is partially or totally oxidized to hematite. No correlation was recognized between the type of iron oxide phenocrysts and rock type. In contrast, the nonphenocrystic iron oxide minerals found in the groundmasses of the volcanic rocks vary with rock type and with the degree of crystallization. These fine-grained iron oxides are believed to be the principal contributors to the remanent magnetization (RM). For this reason a brief discussion of their occurrence in the three principal rock types studied (bedded tuff, welded tuff, and lava) is included.

Most of the bedded tuff studied are believed to be highly lithified deposits of air-fall ash,

[1] Now at U. S. Geological Survey, Federal Center, Denver, Colorado.

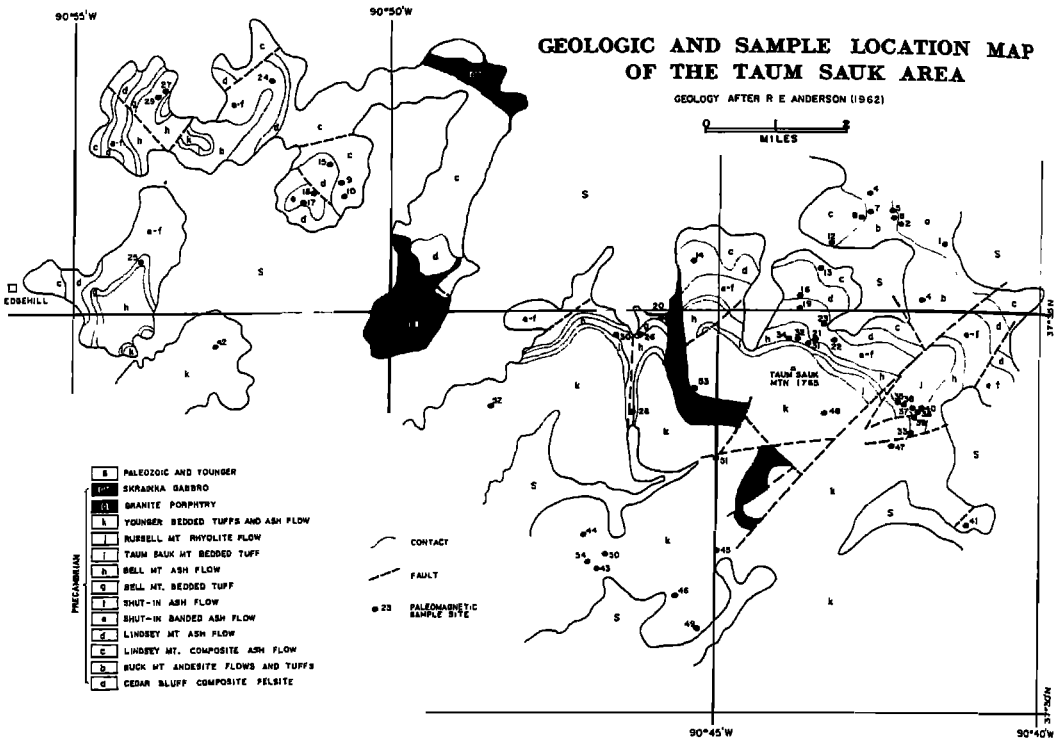


Fig. 1. Map showing spatial relations of volcanic rocks in the Taum Sauk Mountain area, with sampling localities.

lapilli, and crystals that were spread evenly over broad areas. Some beds are persistently red (unit *i* and the bedded tuff of unit *k*) and some are dark gray to black (those in unit *b*). The iron oxide, probably hematite, in the red varieties occurs as finely disseminated dust. Magnetite predominates in the dark gray to black varieties. The rocks are all dense, competent cryptocrystalline felsites, but initially the matrix of these rocks was probably composed largely of unconsolidated vitric ash. It is not known what, if any, changes took place in the iron oxide minerals accompanying the lithification and devitrification of these rocks.

Fine-grained euhedral magnetite is the predominant opaque mineral in the welded ash-flow tuffs which are characterized by cooling-dependent devitrification. This mineral appears to have formed by vapor-phase crystallization and by devitrification. Some welded tuffs show no evidence of these types of crystallization and in them the predominant opaque mineral in the groundmass is very fine-grained hematite. Many rocks are intermediate between these two types.

Reflected light studies of several samples of lava (unit *j*) indicate that hematite is the predominant fine-grained iron oxide. This mineral appears to have formed by vapor-phase crystallization and devitrification.

Sampling and Measurements

The areal distribution of the sample sites is given in Figure 1. In the field the samples were oriented in situ relative to geographic north and the horizontal. In the laboratory they were drilled perpendicular to the horizontal, producing cores 2.5 cm^3 in diameter and 2.3 cm in length. RM measurements were made on cores prepared in this manner from the 54 samples. Five of the samples possessed very low intensity (less than 1×10^{-6} emu/cc) and were difficult to measure.

The measurements were made with the spinner magnetometer in the Rock Magnetic Laboratory of the U. S. Geological Survey in Menlo Park, California. Necessary ac demagnetization was performed in the same laboratory. To determine the proper peak field for partial demag-

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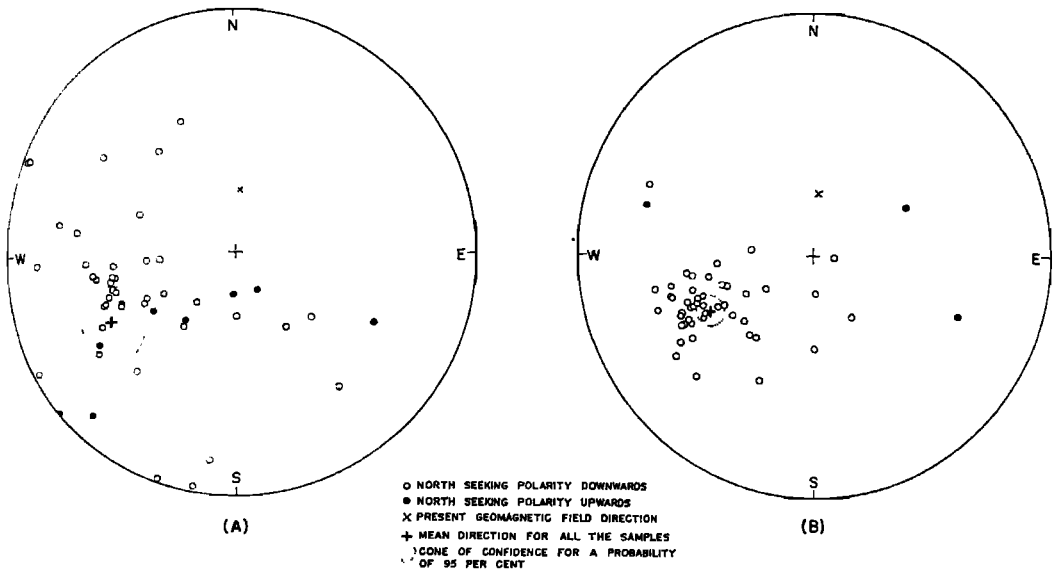


Fig. 2. Equal-area projections on lower hemisphere of the remanent magnetization vectors of all the samples. (*A*) Before partial demagnetization, (*B*) after partial demagnetization in an ac field of 500 to 800 oe.

netization for all the samples, eight samples from different units were demagnitized step by step in increasing ac field strengths from 0 to 600 or 800 oe. Results indicate that most of the RM directions give the best grouping at field strengths between 400 and 600 oe, only the RM directions from unit *j* indicating the requirement of field strengths higher than 800 oe [*Hsu*, 1962]. Therefore, the samples from unit *j* were demagnetized in a peak field of 800 oe and the remaining samples in a peak field of 500 oe.

Direction and Intensity of Remanent Magnetization

Figure 2*A* is a lower-hemisphere equal-area projection of directions of all the samples before demagnetization; Figure 2*B* is a similar plot of directions measured from the samples after demagnetization. The scatter is reduced by removal of the soft component of magnetization. The mean directions of magnetization calculated before and after demagnetization are almost the same, though the intensity is changed. This indicates that the directions of the soft components of magnetization are essentially random. Mean directions of magnetization for individual stratigraphic units are similar (Table 1), but the amount of scatter varies from unit to unit. Units in which several rock types have been combined (*c* and *k*, for example) show wide scatter, whereas homogeneous rock units (*d*, *f*, and *h*, for example) show less scatter. A tilt correction was applied to all the samples which were not horizontal in the field. This correction increased rather than decreased the

TABLE 1. Statistical Analysis of Mean Directions

Unit	*N*	*D*	*I*	*k*	a_{95}
k	10	233.5	51.7	4.2	21.3
j	7	244.4	43.1	10.0	16.7
i	1	247.5	43.7		
h	6	237.7	49.1	20.6	12.4
g	2	249.3	47.8	40.9	15.5
f	5	251.7	44.9	159.6	4.9
e	2	246.6	43.5	206.7	6.9
d	3	256.3	40.8	74.3	9.5
c	6	221.1	50.9	7.1	21.4
b	5	237.7	63.6	13.0	17.4
a	2	243.1	38.8	717.1	3.7
All	49	242.5	48.9	17.2	5.0
All (*T*)	49	243.3	41.0	11.1	6.2

N, number of samples; *D*, declination; *I*, inclination; *k*, precision parameter; a_{95}, confidence angle for a probability of 95%; *T*, after tilt correction.

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scattering of directions of magnetization (Table 1). This feature is in accord with the geologic interpretation of the types of structures involved, none of which is tectonic.

Although the total number of samples is too small to permit a statistically significant comparison between RM intensity or coercivity and stratigraphic position, the data can be evaluated subjectively. Figure 3*A* shows the ratio of the RM intensity before and after demagnetization in a 500-oe ac field for all the samples. This ratio is controlled by coercivity, and therefore the diagram can be regarded as a plot of the relative coercivity of the whole-rock samples. Figure 3*B* shows the intensity of RM after demagnetization. In general, the intensity varies with magnetite content. The relatively high intensity of units *d* and *f* reflects the predominance of magnetite in these rocks, whereas the relatively low intensity of unit *h* reflects the predominance of hematite. Hematite is the only iron oxide mineral recognized in unit *i* and in the bedded tuffs of unit *k*, and the intensities are correspondingly very low; in many cases they are too low to be measured (Figure 3*B*). The relative coercivity (Figure 3*A*) of most of the units is highly variable. This variability probably results, in part at least, from some of the samples having acquired a component of isothermal remanent magnetization through lightning discharges [*Cox*, 1961]. Units *d* and *f* are exceptional in that the relative coercivity is nearly unity in value. This high relative coercivity may result from the abundance of fine-grained magnetite in the groundmass of these rocks, or it may be due to the fact that the magnetite grains contain lamellas and blebs of ilmenite, which would decrease the domain size of magnetite and increase the coercive force.

Some of the variation in the relative coercivity appears to result from an antipolar relationship between the soft and hard components. Individual samples from units *c*, *d*, *e*, *f*, *h*, and *k* show values greater than unity. There is good evidence that this property is real. First, it is characteristic of at least 14 samples. Second, two of these samples, for which there are demagnetization spectrums, show the progressive development of this property. A soft component is presumed to have been removed by demagnetization, leaving a hard component that is more intense than the

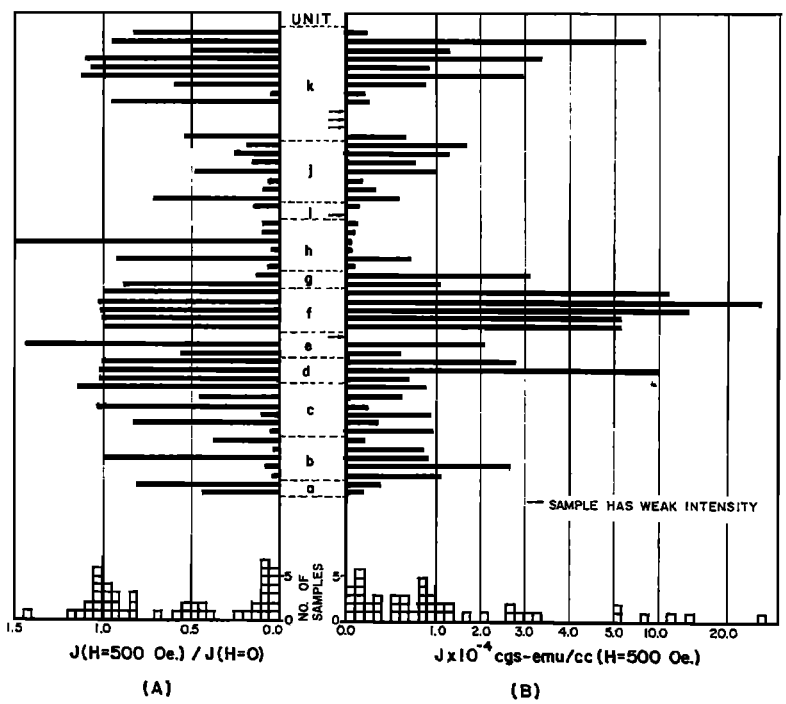


Fig. 3. Intensity variations of the remanent magnetization of all the samples.

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initial remanent magnetization before demagnetization. The accompanying histogram in Figure 3*A* suggests a tri-modal characteristic which may indicate that the hard component of magnetization of all the samples has three different grades of stability.

Origin of Remanent Magnetization

Most of the bedded tuffs studied are believed to be highly lithified deposits of air-fall ash, lapilli, and crystals. The lateral persistence of either hematite or magnetite as the predominant iron oxide in individual beds suggests a depositional rather than postdepositional control. The only type of primary RM that could have been acquired by these rocks is depositional RM (DRM). Usually the intensity of DRM is correspondingly weak [*King*, 1955; *King and Rees*, 1966; *Griffiths et al.*, 1960], but a component of chemical RM (CRM) of these rocks might have resulted from the rudimentary devitrification of the initially vitric matrix or from oxidation accompanying groundwater or hydrothermal alteration. The samples whose RM was too weak to be measured (Figure 3*B*) were found to be very fine-grained, cryptocrystalline bedded tuffs.

Most of the ash-flow tuffs studied are densely welded. It is likely that these rocks and the lavas acquired a component of thermal RM (TRM) during cooling through the Curie temperature of magnetite and/or hematite. As noted above, the iron oxide assemblage present in some of the welded tuffs when they were emplaced was modified or completely reconstituted by cooling-dependent devitrification. In units *d* and *f* and to a lesser degree in units *c* and *k* fine-grained magnetite was formed by vapor-phase crystallization and devitrification. This type of crystallization can be expected over a broad temperature range that may extend as low as 200°C. The RM in these rocks is characteristically stable and relatively intense and is considered to be CRM [*Haigh*, 1958; *Kobayashi*, 1959].

Some of the welded tuffs (unit *h* especially) show little or no evidence of cooling-dependent devitrification or vapor-phase crystallization. The groundmass iron oxide in these rocks is very finely disseminated hematite dust. The hard component RM is generally weak and may be TRM. The data are inconclusive, however.

Pole Position

The mean direction of RM for individual map units examined in this study are similar to the mean direction for all samples combined (Table 1 and Figure 2), which, in turn, is similar to the mean direction established by *Hays and Scharon* [1966] in their broad study of many different Precambrian igneous rock types in the St. Francois Mountains. The mean direction is also similar to the direction determined by *Ku* [1965] for Precambrian volcanic rocks that are known to be both older and younger than the rocks we studied. These directions agree very well with directions determined from Precambrian gabbro [*Hays*, 1961] in which the hard component of RM is almost certainly TRM. The gabbro intrudes the volcanic rocks and is therefore younger. These data indicate that the geomagnetic field was stable throughout this epoch of Precambrian igneous activity and that the CRM inferred in some of the welded tuffs was acquired in response to this field.

Radiometric ages for Precambrian granite from the St. Francois Mountains [*Tilton et al.*, 1962] are about 1300–1400 million years. The mean pole position calculated from the present investigation (Figure 4) is 141.0°W, 0.9°S.

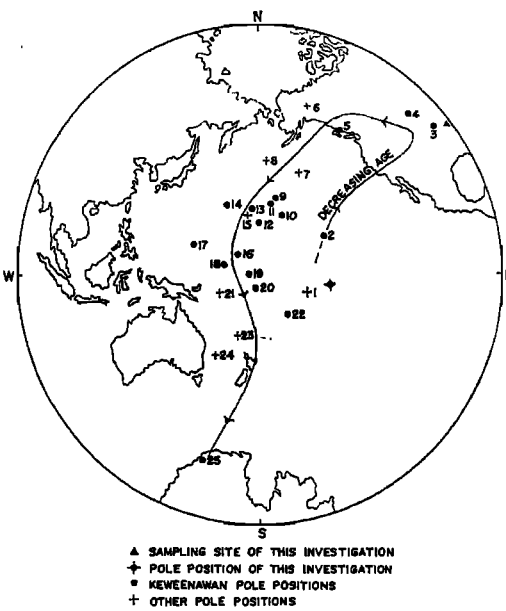


Fig. 4. Late Precambrian pole positions of North America [after *Du Bois*, 1962].

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This pole position fits very well with the late Precambrian polar wandering curve of North America as interpreted by *Du Bois* [1962].

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